CA Lab Practice - Set 3

Hotel Room Bookings

Your task

The web page and the Express server are ready. The database part is missing. Complete it using Mongoose.

- Write your answer in **database-question.js**. The comments in it tell you what to write.
- Do not change any other file.
- The answer is in **database-answer.js**. Try on your own first.

Steps

1. Open this project folder in VS Code and open a terminal in it.
2. Install the packages:

```
npm install
npm install mongoose
```

3. Make sure MongoDB is running, then add the sample data (see below).
4. Complete **database-question.js**.
5. Start the server:

```
node index.js
```

6. Open **http://localhost:3000** in your browser.

After every change to database-question.js, stop the server with Ctrl + C and start it again.

To run the answer instead, start the server with: **node index.js answer**

Database details

Database: hotel

Collection: rooms

Fields: room (text), type (text), floor (number), rent (number)

Sample data

Open mongosh, paste these commands and press Enter. Paste them only once.

```
use hotel
db.rooms.insertMany([
  { room: "101", type: "Single", floor: 1, rent: 1800 },
  { room: "102", type: "Double", floor: 1, rent: 2600 },
  { room: "103", type: "Suite", floor: 1, rent: 5200 },
  { room: "201", type: "Single", floor: 2, rent: 1900 },
  { room: "202", type: "Double", floor: 2, rent: 2800 },
  { room: "203", type: "Suite", floor: 2, rent: 5600 },
  { room: "301", type: "Single", floor: 3, rent: 2100 },
  { room: "302", type: "Double", floor: 3, rent: 3100 },
  { room: "303", type: "Suite", floor: 3, rent: 6100 },
  { room: "401", type: "Single", floor: 4, rent: 2300 },
  { room: "402", type: "Double", floor: 4, rent: 3400 },
  { room: "403", type: "Suite", floor: 4, rent: 6800 }
])
```

Expected output

1. Terminal. Run node index.js.

Expected: Connected to MongoDB, then Server running at http://localhost:3000

2. Show all. Click "Show All Rooms".

Expected: 12 rooms.

3. Search. Choose Suite and click Search.

Expected: 4 rooms: 103, 203, 303, 403.

4. Filter. Type 2700 in "Rent less than" and click Filter.

Expected: 5 rooms: 101, 102, 201, 301, 401.

5. 3 Costliest Rooms. Click "3 Costliest Rooms".

Expected: 403 (6800), 303 (6100), 203 (5600), in this order.

6. Add Room. Fill the form with Room: 501, Type: Suite, Floor: 5, Rent: 7200. Click "Add Room".

Expected: The message "Room saved" appears and the list shows 13 rooms, including 501.